

Alba Cervera-Lierta | Full CV

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Education

Doctor of Philosophy (PhD) in Physics <i>Supervisor: Dr. José Ignacio Latorre</i> Title: Maximal Entanglement: Applications to Quantum Computation and Particle Physics.	Universitat de Barcelona 2015 – 2019
Master in Astrophysics, Particle Physics and Cosmology <i>Specialized in Particle Physics and Gravitation.</i> Thesis title: Non-unitary neutrino oscillations.	Universitat de Barcelona 2014 – 2015
Bachelor in Physics <i>Specialized in Fundamental Physics</i> Thesis title: Scaling of entanglement support for iTEBD algorithm.	Universitat de Barcelona 2009 – 2014

Positions

Research	
Senior Researcher <i>Quantic group (CASE department), Ramón y Cajal fellow since 2024.</i>	Barcelona Supercomputing Center 2021 – Present
Postdoctoral Fellow <i>Alán Aspuru-Guzik group</i>	University of Toronto 2019 – 2021
Technical Researcher <i>Quantic group (CASE department)</i>	Barcelona Supercomputing Center 2017 – 2019
Specialized Technician <i>Project: "Entanglement, tensor networks and cold gases"</i>	Universitat de Barcelona 2014
Coordinator	
Quantum Spain project <i>Project Principal Investigator (budget: 22M €)</i>	Barcelona Supercomputing Center 2021 – Present
Visitor	
Center of Quantum Technologies <i>Prof. Kwek Leong Chuan group.</i>	National University of Singapore May 2017
Instituto de Física Teórica <i>Dr. César Gómez's group.</i>	CSIC Feb 2017
Grupo de investigación en Información y Computación Cuántica <i>Dr. Vicente Martín group</i>	Universidad Politécnica de Madrid Sep 2016
Rudolph Pierls Institute for Theoretical Physics <i>Dr. Juan Rojo's group.</i>	University of Oxford Jan - May 2016

Funded projects

European

EuroQHPC-Integration project **EuroHPC - Joint Undertaking**
Budget: 745.200 €(WP), Role: EuroQCS-Spain consortia coordinator. 2025 – 2029

National

Logic Artificial Intelligence for Photonics (LOGITONICS) **Agencia Estatal de Investigación (AEI)**
Budget: 75k €, Role: principal investigator. 2024 – 2028

Ramón y Cajal fellowship **Agencia Estatal de Investigación (AEI)**
Budget: 50k €, Tenure track position. 2024 – 2029

Quantum Spain project **Secretary of State of Digitalization and AI (SEDIA)**
Budget: 22M €, Role: principal investigator 2021 – 2025
<https://quantumspain-project.es>

Industry

Quantum Experience for Academic Research Program **IBM**
Funding: Exclusive access to pulse-level control of IBM quantum devices, Role: co-PI. 2021 – 2022
Project: High-dimensional quantum experiments superconducting circuits.

Awards and Honors

DonaTIC Revelació **Generalitat de Catalunya**
October 2025

DTIC - Ingenierías UPF research award for women researchers **Universitat Pompeu Fabra**
May 2024

PhD Excellent Cum Laude **Universitat de Barcelona**
Jun 2019

Teach Me QISKit Award **IBM**
Jun 2018

Publications

Link to [Google Scholar](#) and [Orcid](#).

24. Towards interpretable AI with quantum annealing feature selection **Preprint**
F. Aldo Venturelli, E. Costa, Sikha O K, B. Juliá-Díaz, M. A. González Ballester, A. Cervera-Liarta 2026
arXiv:2604.25649 [cs.LG].

23. Warm Starts, Cold States: Exploiting Adiabaticity for Variational Ground-States **Preprint**
R. Puig, B. Casas, A. Cervera-Liarta, Z. Holmes, A. Pérez-Salinas 2026
arXiv:2602.06137 [quant-ph].

22. Hardware-inspired Continuous Variables Quantum Optical Neural Networks **Preprint**
T. Krasimirov-Ivanov, A. Cervera-Liarta, P. Stornati, F. Centrone 2025
arXiv:2512.05204 [quant-ph].

21. Gauge invariance from quantum information principles **Preprint**
C. Núñez, M. Pardina, M. Asorey, J. I. Latorre, A. Cervera-Liarta 2025
arXiv:2511.04358 [hep-th].

20. Orchestrating Quantum-HPC Workflows with Distributed Quantum Circuit Cutting **SC Workshops'25**
2025
M. Tejedor, B. Casas, F. Conejero, A. Cervera-Liarta, R. Badia

Proceedings of the SC'25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis, 1898 - 1906.

- 19. Universality of entanglement in gluon dynamics** SciPost Phys.
C. Núñez, A. Cervera-Liarta, J. I. Latorre
SciPost Physics **20**, 063 (2026). 2026
- 18. Quantum Circuits for High-Dimensional Absolutely Maximally Entangled States** QST
B. Casas, G. Rajchel-Mieldzioć, S. Ahmad Rather, M. Płodzień, W. Bruzda,
A. Cervera-Liarta, K. Życzkowski, Quantum Science and Technology **11**(2), 025011. 2026
- 17. Multidimensional Fourier series with quantum circuits** Phys. Rev. A
B. Casas, A. Cervera-Liarta
Physical Review A **107**, 062612 (2023). 2023
- 16. Design of quantum optical experiments with Logic Artificial Intelligence** Quantum
A. Cervera-Liarta, M. Krenn, A. Aspuru-Guzik
Quantum **6**, 836 (2022). 2022
- 15. On scientific understanding with artificial intelligence** Nat. Rev. Phys.
M. Krenn, R. Pollice, S. Y. Guo, et. al.
Nature Review Physics (2022). 2022
- 14. Toward Reliability in the NISQ Era: Robust Interval Guarantee for Quantum Measurements on Approximate States** Phys. Rev. Research
M. Weber, A. Anand, A. Cervera-Liarta, J. S. Kottmann, T. H. Kyaw, et. al.
Physical Review Research **4**, 033217 (2022). 2022
- 13. Programming Physical Quantum Systems with Pulse-Level Control** Front. Phys.
K. N. Smith, G. S. Ravi, T. Alexander, et. al.
Frontiers in Physics, **672** (2022). 2022
- 12. Learning Interpretable Representations of Entanglement in Quantum Optics Experiments using Deep Generative Models** Nat. Mach. Intell.
D. Flam-Shepherd, T. Wu, X. Gu, A. Cervera-Liarta, M. Krenn, A. Aspuru-Guzik
Nature Machine Intelligence **4** (6), 544-554 (2022). 2021
- 11. Experimental high-dimensional Greenberger–Horne–Zeilinger entanglement with superconducting transmon qutrits** Phys. Rev. Appl.
A. Cervera-Liarta, M. Krenn, A. Aspuru-Guzik, A. Galda
Physical Review Applied **17**, 024062 (2022). 2022
- 10. Noisy intermediate-scale quantum algorithms** Rev. Mod. Phys.
K. Bharti, A. Cervera-Liarta, T. H. Kyaw, T. Haug, et. al.
Review of Modern Physics **94**, 015004 (2022). 2022
- 9. The Meta-Variational Quantum Eigensolver (Meta-VQE): Learning energy profiles of parameterized Hamiltonians for quantum simulation** PRX Quantum
A. Cervera-Liarta, J. S. Kottmann and A. Aspuru-Guzik
PRX Quantum **2**, 020329 (2021). 2021
- 8. Tequila: A platform for rapid development of quantum algorithms** Quantum Sci. Technol.
J. S. Kottmann, S. Alperin-Lea, T. Tamayo-Mendoza, A. Cervera-Liarta et. al.
Quantum Science and Technology **6** 024009 (2021). 2021
- 7. Data Re-uploading for a Universal Quantum Classifier** Quantum
A. Pérez-Salinas, A. Cervera-Liarta, E. Gil-Fuster, J. I. Latorre
Quantum **4**, 226 (2020). 2020
- 6. Maximal Entanglement: Applications in Quantum Information and Particle Physics** PhD Thesis
A. Cervera-Liarta
arXiv:1906.12099 [quant-ph]. 2019
- 5. Quantum Circuits for Maximally Entangled States** Phys. Rev. A
A. Cervera-Liarta, J. I. Latorre, D. Goyeneche
Physical Review A **100**, 022342 (2019). 2019
- 4. Exact Ising Simulation on a Quantum Computer** Quantum
A. Cervera-Liarta
Quantum **2**, 114 (2018). 2018

- 3. Multipartite Entanglement in Spin Chains and the Hyperdeterminant** J. Phys. A Math. Theor. 2018
A. Cervera-Lierta, A. Gasull, J. I. Latorre and G. Sierra.
 Journal of Physics A: Mathematical and Theoretical **51**, 505301 (2018).
- 2. Maximal Entanglement in High Energy Physics** SciPost Phys. 2017
A. Cervera-Lierta, J. I. Latorre, J. Rojo and L. Rottoli
 SciPost Physics **3**, 036 (2017).
- 1. Operational Approach to Bell Inequalities: Application to qutrits** Phys. Rev. A 2016
D. Alsina, A. Cervera, D. Goyeneche, J. I. Latorre and K. Zyczkowski
 Physical Review A **94**, 032102 (2016).

Reports and white papers.....

- When matter and information merge into “Quantum”** Commun. Phys. 2023
R. Aguado, A. Cervera-Lierta, A. Correia, et. al., Communications Physics **6**(1) 266 (2023).
- Workflows Community Summit 2022: A Roadmap Revolution** Technical Report 2023
R. Ferreira da Silva, R. M. Badia, V. Bala, et. al.
 ORNL/TM-2023/2885, arXiv:2304.00019 [cs.DC].

Books, chapters and notes.....

- Machine learning in quantum sciences** Cambridge University Press 2025
A. Dawid, J. Arnold, B. Requena, et. al.
 Cambridge University Press 2025, arXiv:2204.04198 [quant-ph].
- Simulation of the 1d XY model on a quantum computer** SciPost Phys. Lect. Notes 2025
M. Farreras, A. Cervera-Lierta
 SciPost Physics Lecture Notes **95** (2025).

Software development.....

Libraries:

- Distributed quantum circuit cutting library “Qdislib”** Python 2025
Project collaborator, <https://qdislib.readthedocs.io/>
- Quantum computing open-source library “Tequila”** Python 2021
Project collaborator, <https://github.com/tequilahub/>
- Codes and demos:

- Logic AI algorithm “Klaus”** Mathematica 2022
Creator and developer, <https://github.com/AlbaCL/Klaus>
- Meta Variational Quantum Eigensolver** Python 2021
Co-creator and developer., <https://github.com/aspuru-guzik-group/Meta-VQE>

Talks and seminars

Some of my talks' videos are listed in my [Youtube Channel playlists](#).
 Virtual talks due to COVID-19 pandemic.

Keynote.....

- Fusion HPC workshop** Virtual Feb 24, 2026
“Quantum-HPC integration: state-of-the-art, challenges and opportunities”
- Lliçó Innaugural** UPF Sep 30 2024
“Tecnologies quàntiques: passat, present i futur d’una (r)evolució tecnològica”
 Enginyeries Universitat Pompeu Fabra.
- 100tífiques** UPC Feb 1 2024
“Tecnologies quàntiques: passat, present i futur d’una (r)evolució tecnològica”
 Fundació Catalana per a la Recerca i la Innovació.
- The quest for the quantum computer** 10th BSC Doctoral Symposium May 9, 2023
Barcelona, Spain

The Noisy Intermediate-Scale Quantum era

Porto Alegre, Brasil

CARLA 2022

Sep 28-30, 2022

Margarita Salas Honoris Causa PhD act

"De las computadoras a los ordenadores"

Universidad de Extremadura

Feb 2 2019

Colloquium.....

Computación cuántica: pasado, presente y futuro

de una revolución tecnológica

Opening academic year Aula de Ciencia y Tecnología, Universidad de Alicante, Spain.

Universidad de Alicante

Sep 28, 2023

Tecnologies quàntiques: present i futur

Universitat de Barcelona, Spain.

ICCUB Colloquia

Feb 16, 2022

Invited.....

Quantum Spain

QuantumMatter'26

Barcelona, Spain.

Apr 27, 2026

Gauge Invariance from quantum information principles

Quantum Observables for Collider Physics Workshop.

CERN

Apr 23, 2026

A quantum computer decalogue

XL Trobades de la Mediterrània

Maó, Spain.

Oct 15, 2025

Quantum-HPC integration: challenges and opportunities

SQA 2025

Delft, The Netherlands.

Aug 26, 2025

Quantum computers: the new supercomputer partition

HiPEAC 2025, Classical HPC & QC: the way to foster the integration workshop

Barcelona, Spain.

Jan 20, 2025

Quantum computers: the new supercomputer partition

SC24

Atlanta, USA

Nov 20, 2024

Design of quantum optical experiments with logic artificial intelligence

SCOP24

ICFO, Barcelona

Oct 3, 2024

Building the Spanish Quantum Computer

Quantum Technologies in Europe, Madrid, Spain

Quantera

Nov 23, 2023

Quantum-HPC integration: challenges and opportunities

EQTC 2023

Hannover, Germany

Oct 19, 2023

Variational Quantum Algorithms

QTYR 2023

CSIC, Madrid

Jul 04, 2023

Quantum Spain in the NISQ era

QBN Meeting on Quantum Computing and Applications

BSC-CNS

Jun 15, 2023

High-dimensional quantum computation: opportunities and challenges

Symposium on Open-Source Tools for Quantum Computing & Simulation

ICFO

Mar 31, 2023

High-dimensional quantum computation

CERN

QT4HPC

Nov 2, 2022

Quantum algorithms in near term devices

IQTC, University of Barcelona (Spain).

New Trends in Computational Chemistry

Sep 9, 2022

Quantum Spain

Barcelona (Spain).

Quantum 2022

Jun 22, 2022

Quantum Spain: un impulso para el ecosistema de computación cuántica

NASERTIC, Pamplona (Spain).

Jornada cuántica

May 4, 2022

Quantum Spain: un impulso para el ecosistema de computación cuántica

SCAYLE, León (Spain).

Jornada Eco. Cuántico

Mar 1, 2022

Near term quantum computation: opportunities and challenges

Bilbao (Spain).

Quantum 2021

Nov 24, 2021

Noisy Intermediate-Scale Quantum era <i>Virtual (*)</i> .	JURES 2021 <i>Sep 17, 2021</i>
Noisy Intermediate-Scale Quantum algorithms <i>Perspectives on Quantum Sensing and Computation for Particle Physics. Virtual (*)</i> .	CERN <i>Jul 15, 2021</i>
The Meta-Variational Quantum Eigensolver <i>Virtual (*)</i> .	Machine Learning for Quantum X <i>Jul 6, 2021</i>
The Meta-Variational Quantum Eigensolver <i>Virtual (*)</i> .	QWorld Quantum Science Days <i>Jun 1, 2021</i>
Chronicles of Entangland, the foundation of the SC town <i>Virtual (*)</i> .	Qiskit Hackathon Europe <i>May 26, 2021</i>
Experimental high-dimensional GHZ entanglement with superconducting transmon qutrits <i>Chicago. Virtual (*)</i> .	Pulse-level Quantum Control Workshop <i>May 18, 2021</i>
Maximal Entanglement in High Energy Physics <i>Topical Group on Hadronic Physics. Virtual (*)</i> .	APS April Meeting <i>Apr 15, 2021</i>
Noisy Intermediate-Scale Quantum algorithms <i>Virtual (*)</i> .	Universidad Iberoamericana <i>Apr 13, 2021</i>
Please, feed the quantum troll! <i>Virtual (*)</i> .	Q-Hack 2021 <i>Feb 19, 2021</i>
Testing and designing quantum algorithms for the NISQ era <i>Quantum Computing and Simulation on Near-Term Devices. Virtual (*)</i> .	INT-20-3 <i>Nov 21, 2020</i>
Maximal Entanglement in High Energy Physics, Quantum Entanglement at Collider Energies <i>Stony Brook (United States)</i> .	CFNS Stony Brook <i>Sep 11, 2018</i>
Exact simulation of a Quantum Phase Transition on a quantum computer <i>Barcelona (Spain)</i> .	Quantum Phase Transitions <i>Apr 05, 2018</i>
Emergence of Gauge Symmetry from a Maximal Entanglement Principle <i>Madrid (Spain)</i> .	Universidad Complutense de Madrid <i>Sep 21, 2016</i>
Contributed	
The Meta-Variational Quantum Eigensolver <i>Virtual (*)</i> .	APS March Meeting 2021 <i>Mar 18, 2021</i>
Data re-uploading for a universal quantum classifier <i>Virtual (*)</i> .	Q-Turn 2020 <i>Nov 24, 2020</i>
Data re-uploading for a universal quantum classifier <i>Quantum Techniques Machine Learning. Virtual (*)</i> .	QTML 2020 <i>Nov 12, 2020</i>
Maximal Entanglement in Quantum Computation <i>Barcelona (Spain)</i> .	BSC Severo Ochoa Doctoral Symposium <i>May 07, 2019</i>
Seminars	
Ordenadores cuánticos: la nueva partición de los superordenadores <i>Ciclo de charlas por el Año Internacional de la Ciencia y Tecnología Cuántica</i> .	Universidad de Sevilla <i>May 14, 2025</i>
Computació quàntica: reptes i oportunitats <i>FIB</i>	UPC <i>Mar 11, 2025</i>
The Noisy Intermediate-Scale Quantum Computation <i>IQCC</i>	Universitat de Girona <i>Jan 21, 2025</i>
Computación cuántica: pasado, presente y futuro de una revolución tecnológica <i>Madrid, Spain</i>	Telefónica <i>Feb 21, 2024</i>

Quantum-HPC integration: challenges and opportunities <i>Toronto, Canada</i>	LG Toronto <i>Oct 26, 2023</i>
Quantum Spain: creating a public quantum-HPC infrastructure <i>European Commission (Virtual).</i>	Quantum Cafe <i>Sep 21, 2023</i>
A roadmap to quantum computing integration into HPC infrastructures	RISC2 Seminar Series <i>Mar 15, 2023</i>
Noisy Intermediate Scale Quantum Computation <i>Dept. Condensed Matter Physics.</i>	Universidad de Zaragoza <i>Jun 3, 2022</i>
Noisy Intermediate Scale Quantum Computation <i>Virtual.</i>	Universidad Nacional de Colombia <i>Apr 21, 2022</i>
Design of quantum optical experiments with logic artificial intelligence <i>Quantum Optics group seminar.</i>	MPQ <i>Oct 29, 2021</i>
High-dimensional quantum computing experiments <i>Prof. Manas Mukherjee group meeting (Singapore, Virtual *)</i>	CQT <i>Aug 03, 2021</i>
Maximal Entanglement in a Quantum Computer <i>Fermilab theory seminars. Virtual (*)</i>	Fermilab <i>Jun 15, 2021</i>
The Meta-Variational Quantum Eigensolver <i>Oxford Quantum Information Society. Virtual (*)</i>	Oxford <i>May 20, 2021</i>
The Meta-Variational Quantum Eigensolver <i>Virtual.</i>	Scientific Machine Learning Webinars <i>May 6, 2021</i>
Noisy Intermediate-Scale Quantum algorithms <i>Hypatia Series. Virtual(*)</i>	Alexandria University <i>Apr 8, 2021</i>
Maximal Entanglement in High Energy Physics <i>Nuclear Theory series. Virtual(*)</i>	University of Maryland <i>Apr 2, 2021</i>
Noisy Intermediate-Scale Quantum algorithms <i>Belgian Quantum Physics Initiative. Virtual(*)</i>	Académie Royale de Belgique <i>Mar 4, 2021</i>
Designing VQA with Tequila <i>Virtual.</i>	Quantum Conversations by the Bay <i>Nov 28, 2020</i>
Maximal Entanglement. Applications to Quantum Computation and Particle Physics <i>Matterlab group meeting. Toronto (Canada).</i>	UofT <i>Mar 1, 2029</i>

Panels and round tables

As a moderator.....

Round table Industrial forum <i>Participants: Marta Estarellas, Richard Kienhoefer, Pascale Senellart, Fernando de la Iglesia and Bilal Kalyoncu</i>	Quantum Matter 2023 <i>May 24, 2023</i>
Round table Industrial forum <i>Participants: Marc Almendros, Niels Bultink, Gianni del Bimbo, Daniel Szombati, Yonatan Cohen, Josep M. Martorell, Heike Riel and Carmen Palacios.</i>	Quantum Matter 2022 <i>Jun 22, 2022</i>

As a participant.....

Quantum computing + Communications: meet at the halfway. <i>Pannel with Hugues de Riedmatten.</i> <i>Modarated by Carmen G. Almudéver.</i>	PuzzleX <i>Oct 7, 2022</i>
Catalan Nanotech Ecosystem Session <i>Modarated by Stephan Roche.</i>	ACM Nanocomm <i>Oct 7, 2022</i>
Women in ACM/IEEE WICE Luncheon <i>Modarated by Núria Castell.</i>	ACM Nanocomm <i>Oct 6, 2022</i>

Quantum Computing: Changing the Course of Industry
Round table with Carlos Kuchkovsky, Marta Pita, and Enrique Sanchez.
Moderated by Jaime Gómez.

Barcelona New Economy Week
Oct 3, 2022

Quantum Technologies
Panel with Silvia Carrasco, Carlos Abellán and Enrique Mora.
Moderated by Rubén Criado

Barcelona Deep Tech Summit
Sep 22, 2022

The Impact of Advanced Computing in AI
Round table with Jean Pierre Panziera and Nicolas Moes.
Moderated by David Pringle

Science|Business
Sep 19 2022

The Future of HPC
Panel with Paul Messina, Per Stenström, Uri Weiser and Min Li.
Moderated by Josephine Wood

ACM Europe Summer School
Sep 2 2022

Young People Program Career Panel

DATE 2022
Mar 21 2022

Quantum Software Ecosystem
Pannel discussion with industry and academic members

IEEE Services
Sep 9 2021

Committees and referee

National

Vice-president
Grupo Especializado en Información y Tecnologías Cuánticas (GEITC).
Real Sociedad Española de Física
2023 - Present

European

Quantum Computing working group
Quantum Coordination Board
European Quantum Flagship and EuroHPC
2023 - Present

Quantum Standarization working group
Quantum Coordination Board
European Quantum Flagship and EuroHPC
2023 - Present

Quantum - HPC Working group
European Quantum Flagship
2022 - Present

Scientific Advisory Board

Finish Quantum Initiative
CSC, Aalto University, VTT
2023 - 2026

Consell de les Tecnologies Quàntiques de Catalunya
Board member
Generalitat de Catalunya
2022 - 2024

Scientific Committee

Talent Arena (Mobile World Congress) 2026, Barcelona.
Quantum Techniques in Machine Learning 2025, Singapore.
Quantum Techniques in Machine Learning 2023, CERN.
ICE8 2023, Santiago de Compostela.
Quantum Matter 2023, Madrid.
Quantum Matter 2022, Barcelona.
Quantum Techniques in Machine Learning 2020, Virtual.

Peer-review and referee

Journals: Nature Communications, Nature Machine Intelligence, Physical Review A, Advanced Quantum Technologies, Quantum, New Journal of Physics, Quantum Information Processing.

Project agencies: QuantERA, Agencia Española de Investigación.

Access committee: CESGA quantum computing access.

PhD Thesis committees

Quantum-inspired Machine Learning for HEP Applications <i>Ema Puljak</i> Supervisor: Artur Garcia	Universitat Autònoma de Barcelona <i>Jan 23, 2026</i>
Quantum Adventures From Analog to Digital: Gauge, Scars and Laughlin <i>Barbara Andrade Dos Santos</i> Supervisors: Maciej Lewenstein, Tobias Graß	U. Politècnica de Catalunya <i>May 19, 2025</i>
Memory Management in Software Quantum Computing Simulators <i>Gilberto Díaz</i> Supervisors: C. Jaime Barrios Hernández, L. Angelo Steffenel, J. F. Couturier	U. Industrial de Santander <i>Dec 9, 2024</i>
Quantum many-body ground states via digital quantum simulation <i>Bruno Rodrigues Pacheco e Murta</i> Supervisors: Joaquín Fernández-Rossier, Nuno Miguel Machado Reis Peres	Universidade do Minho <i>Feb 29, 2024</i>
Reservoir Computing in Quantum Systems <i>Rodrigo Martínez Peña</i> Supervisor: Roberta Zambrini, Miguel Cornelles Soriano	Universitat de les Illes Balears <i>Sep 7, 2023</i>
Fault tolerant quantum algorithms <i>Pablo Moreno-Casares</i> Supervisor: Miguel Ángel Martín-Delgado	Universidad Complutense de Madrid <i>Jan 17, 2023</i>
Quantum information in lattices <i>Bruna Gabrielly De Moraes Araújo</i> Supervisors: Antonio Acín dal Maschio, Daniel Cavalcanti Santos and Stephan Jean Louis Roche	Universitat Autònoma de Barcelona <i>Jul 22, 2022</i>

Conference and seminars organization

Spring School on Near-Term Quantum Computing II <i>Co-organizer. Centro de Ciencias de Benasque Pedro Pascual.</i>	Benasque, Spain. <i>Mar 15-21, 2026</i>
España Cuántica <i>Co-organizer and chair., Fundación Ramón Areces.</i>	Madrid, Spain. <i>Nov 27-28, 2025</i>
Spring School on Near-Term Quantum Computing I <i>Co-organizer. Centro de Ciencias de Benasque Pedro Pascual.</i>	Benasque, Spain. <i>Apr 14-27, 2024</i>
Quantum Matter International Conference <i>Industrial forum co-organizer.</i>	Madrid <i>May 22-25, 2023</i>
Quantum Research Seminars Toronto. <i>Co-organizer and chair. Youtube: Quantum Research Seminars Toronto.</i>	Virtual <i>2020 – 2021</i>

Teaching

Lectures and courses

A gentle introduction to quantum computing <i>ACM Europe Summer School on HPC</i> The quest for the quantum computer	ACM Europe, Barcelona <i>July 7, 2023</i>
A gentle introduction to quantum computing <i>ACM Europe Summer School on HPC</i>	ACM Europe, Barcelona <i>July 7, 2023</i>
Quantum Algorithms I and II <i>Spring School on Superconducting Qubit Technology</i>	Centro de Ciencias de Benasque Pedro Pascual <i>Apr 11-21, 2023</i>
The Noisy Intermediate Scale Quantum Computation <i>Escenarios Científicos</i>	Universidad Complutense de Madrid <i>Mar 14, 2023</i>
The Noisy Intermediate Scale Quantum Computation <i>Master Modelling for Science and Engineering Research and Innovation</i>	Universitat Autònoma de Barcelona <i>Oct 4, 2022</i>

Variational Quantum Computing <i>IV Summer School on Quantum Computing</i>	Universidad Internacional Menéndez-Pelayo Sep 5-9, 2022
Near-term quantum computing <i>Clase introducción al curso en Quantum Computing software.</i>	Universidad Castilla La-Mancha Jun 20, 2022
NISQ computing <i>Theory lectures</i>	Institut de Ciències Fotòniques (ICFO) Mar 15-24, 2022
Variational Quantum Computing <i>III Summer School on Quantum and Q-Inspired Computing</i>	Universidad Internacional Menéndez-Pelayo Sep 7, 2021
Noisy Intermediate-Scale Quantum Algorithms <i>Summer School "Machine Learning in Quantum Physics and Chemistry"</i>	Warsaw, Poland Aug 25, 2021
Quantum Computing in the NISQ era <i>Talk in Quantum Technologies Winter School</i>	UCL Dec 10, 2020
Bojos per la Supercomputació <i>Introductory course in quantum computation for high-school students</i>	BSC 26 Apr – 04 May, 2019
The Quantum Revolution <i>Seminar at Master in Finance</i>	ESADE Business School Jun 07, 2018

Teaching positions

Teacher Assistant <i>Calculus I-II, Theoretical Mechanics, Physics I - II.</i>	Universitat Autònoma de Barcelona 2017 – 2019
Teacher Assistant <i>Modern Physics Laboratory, Computational Physics.</i>	Universitat de Barcelona 2016 – 2017

Students

PhD

César Fernández Lamas <i>Co-supervised with Javier Argüello Luengo and Antonio Rubio Abadal</i>	Universitat de Barcelona Feb 2026 – Present
Todor Krasmirov	Universitat de Barcelona Oct 2025 – Present
Francesco Aldo Venturelli <i>Co-supervised with Miguel Ángel González Ballester</i>	Universitat Pompeu Fabra Nov 2024 – Present
Berta Casas Font <i>Co-supervised with Artur Garcia</i>	University of Barcelona Sep 2022 – Present

Master

Blanca Belmonte <i>Particle detection using entanglement</i> Co-supervised with Diego Blas. Master Quantum Science and Technology.	University of Barcelona Feb 2026 – Jul 2026
Mar Tejedor Ninou <i>Distributed Circuit Cutting for Hybrid Quantum-HPC Systems</i> Co-supervised with Rosa Badia. Master Quantum Science and Technology.	University of Barcelona Feb 2025 – Jul 2025
Juan Francisco Martín Bravo <i>Quantum benchmarking: entanglement measures in a quantum computer</i> Master Quantum Science and Technology.	University of Barcelona Feb 2023 – Jul 2023
Berta Casas Font <i>Quantum function fitting and classification beyond the single-qubit model</i> Master Quantum Science and Technology.	University of Barcelona Feb 2022 – Jul 2022

Internships

Francesc Sabater Garcia

Adding Qibo as a Supported Framework for Mitiq
Master Quantum Science and Technology.

University of Barcelona

Sep 2023 – Feb 2024

Undergraduate (degree)

Alba Guasch

Fault Tolerant Quantum Computation
Physics Degree. Co-supervised with Sofyan Iblisdir.

Universitat de Barcelona

Sep 2025 – Jan 2026

Natasha Trojan

Optimization of quantum circuit cutting for distributed computing
Computer Science Degree. Co-supervised with Rosa Badia.

Universitat Politècnica de Catalunya

Sep 2025 – Jan 2026

Marc Farreras

Exact XY model simulation with a quantum computer
Physics Degree

University of Barcelona

Feb 2023 – Jul 2023

Alex Alemany

The HHL algorithm
Computer Science Degree. Co-supervised with Jordi Delgado.

Universitat Politècnica de Catalunya

Feb 2023 – Jul 2023

Júlia Barberà Rodríguez

Application of Grover's quantum algorithm for string matching
Physics Degree

University of Barcelona

Feb 2022 – Jul 2022

Elies M. Gil Fuster

Variational Quantum Classifier
Physics Degree. Co-supervised with J. I. Latorre.

University of Barcelona

Feb 2019 – Jul 2019

Outreach

Some of my talks and events videos are listed in my [Youtube Channel](#) playlists.

Articles

Carta a mi yo del futuro

TELOS 129

Fundación Telefónica

May 2025

Computación Cuántica: Una nueva (r)evolución en computación

BIT 236, Especial Tecnologías Cuánticas

BIT

Jun 2025

Computación cuántica: 25 años después

Revista Española de Física 39 (1), 12-16

Revista Española de Física

Jun 2025

**Supercomputación y computación cuántica:
el camino hacia la integración**

Revista Española de Física 38 (2), 37-41

Revista Española de Física

Mar 2024

L'ordinador quàntic: un camí de ciència, tecnologia i sobirania

Revista IDEES núm. 61 "Quantum: una revolució que canviarà el món?".
Translated to Spanish, English and Catalan.

Revista IDEES

Feb 2024

Computación Cuántica en la Nube

Revista Española de Física 34 (1), 25-29.

Revista Española de Física

Mar 2020

How I became a quantum computation scientist

Medium article for @qiskit

Jul 2018

Contributing Writer

Medium articles, @quantum_wa.

Quantum World Association

2018 – 2019

Commissary

City and Science Biennial

With Toni Pou and Teresa Badia.

Ajuntament de Barcelona

2025

Panels and round tables

Universitat de Barcelona. <i>Com entendre, i divulgar, la física quàntica</i> With Dani Jiménez, Oriol Junqueras (UAB), Amílcar Labarta (UB) i Xavier Batlle (UB).	Barcelona Apr 15, 2026
Talent Arena MWC <i>Women influencing and leading the forefront of tech</i> With Marjorie Grassler (MWC), Marta Portalés Oliva (MWC) and Estel Blay (IEEC).	Barcelona Mar 3, 2026
MWC <i>The New Sovereignty: Quantum Power and the redesign of the Global Technological Map</i> With Marta Estarellas (Qilimanjaro Quantum Tech), Vanesa Díaz (LuxQuanta), Silvia Carrasco (ICFO) and Alberto de Vicente (Red.es).	Barcelona Mar 2, 2026
Talent Arena MWC <i>Conversation with Antonio Acín</i>	Barcelona Mar 5, 2025
Jornades REACT <i>Round table with Teresa Riesgo Alcaide, Laia Bonet, Xavier Azemar Mallard Mateo Valero Cortes.</i> Moderated by Elena Arrieta Palacio.	Barcelona Mar 29, 2023
Premios Nacionales de Investigación <i>Presenter and moderator of the act.</i> Chaired by HRH King and Queen of Spain, Spanish Minister of Science and Universities and Barcelona Mayor.	Barcelona May 5, 2022
Mujeres y Digitalización: Por una España Igualitaria <i>Round table with Marta Villegas, Lucía Flecha and Susana Fernández.</i> Moderated by the Secretary of State of Digitalization and AI Carme Artigas.	MINECO Mar 8, 2022
Dia Internacional Dona i Nena a la ciència <i>Event with Barcelona city Mayor and government.</i>	Ajuntament de Barcelona Feb 11, 2022
Women in Physics path <i>Organized by the Equality commission from the Physics Faculty</i>	Universitat de Barcelona Mar 3 2021
Keynote speech	
Bojos de la Ciència <i>Speech about Bojos per la Supercomputació program</i>	Fundació Catalunya-La Pedrera Dec 10 2018
Bojos de la Ciència <i>Speech at Master grants act</i>	Fundació Catalunya-La Pedrera Oct 16 2018
Talks and courses	
Computació quàntica: passat, present i futur <i>Organized by MoraBanc, Andorra Telecom, FEDA, RTVA i Agora International School.</i>	Andorra May 13, 2026
From Particles to Processors: My Journey into Quantum Computing <i>LeaderSHE Talk</i>	ICN2 Apr 8, 2026
Computació quàntica: passat, present i futur d'una (r)evolució tecnològica <i>Mobile World Congress</i>	Beat Barcelona Mar 6, 2025
Quantum computers: the new supercomputer partition <i>Mobile World Congress</i>	BSC Booth. Mar 6, 2025
Computación cuántica: pasado, presente y futuro de una (r)evolución tecnológica <i>Casa del Cordón</i>	Burgos, Spain. Feb 27, 2025
Computació quàntica: passat, present i futur d'una (r)evolució tecnològica <i>Cicle Últimes fronteres: la ciència del segle XXI</i>	Universitat de Girona May 4, 2023
Computación cuántica: presente y futuro de una revolución tecn. <i>Jornada tecnológica.</i>	Uni. Las Palmas de Gran Canaria Apr 18, 2023
Computació Quàntica i els reptes de Talent TIC en els propers anys <i>Col·legi d'Informàtica de Catalunya</i>	Lul·liana Nov 24, 2022

Tecnologies quàntiques: present i futur d'una (r)evolució tecnològica <i>Casal de Joves Sant Just Desvern</i>	Jornades Feministes <i>Nov 20, 2022</i>
Present i futur de les tecnologies quàntiques <i>Two-day introductory course for senior citizens.</i>	A-12 Estudis i Recerca Universitaria <i>Oct 20-26, 2022</i>
Computació quàntica: present i futur d'una revolució tecnològica <i>Universitat Politècnica de Catalunya.</i>	Open Day TIC <i>May 12, 2022</i>
Tecnologías Cuánticas: presente y futuro <i>Barcelona.</i>	Pint of Science <i>May 11, 2022</i>
Quantum Spain: un impulso para el ecosistema de computación cuántica <i>Seminar for the Corporación Tecnológica de Andalucía. Virtual.</i>	CTA <i>Apr 26, 2022</i>
Supercomputación y Quantum Spain <i>Seminar for the Plataforma Tecnológica Española y de Tecnologías Disruptivas. Virtual.</i>	PTE Disruptive <i>Mar 22, 2022</i>
Tecnologías cuánticas: presente y futuro <i>Virtual.</i>	Col · legi de Biòlegs de Catalunya <i>Mar 16, 2022</i>
Computación Cuántica <i>Cosmocaixa (Barcelona)</i>	Live Talks Cosmocaixa <i>Feb 16, 2022</i>
Quantum Spain <i>Quantum Barcelona and Madrid meet-up.</i>	Meet-up <i>Dec 13, 2021/Feb 9, 2022</i>
Quantum Calling <i>Waterloo Quantum Meet-up</i>	K-W Quantum Enthusiasts Meetup <i>Jun 18, 2020</i>
Quantica <i>Quantum computing talk and participation in the exhibition (video)</i>	CCCB <i>Apr 10-Sep 26, 2019</i>
De las computadoras a los ordenadores <i>Badajoz, Spain</i>	Margarita Salas Honoris Causa PhD event <i>Feb 26, 2019</i>
Technology to manipulate Nature <i>Madrid, Spain</i>	El País con tu Futuro <i>Dec 18, 2018</i>
Quantum Computation: a new paradigm <i>Sant Feliu de Llobregat, Barcelona, Spain</i>	TEDx talk <i>Sep 15, 2018</i>
Quantum Computation <i>Researcher's Night 2018</i>	Cosmocaixa (Science Museum) <i>Sep 28, 2018</i>
Mentorship programs.....	
Quantum Open Source Foundation mentorship program <i>Mentor</i>	QOSF <i>Sep 2020 – Jan 2021</i>
Girls E-Mentorship program <i>Mentor</i>	GEM <i>Sep 2020 – June 2021</i>